

VR-Based Kinematic Assessments: Examining the Effects of Task Properties on Arm Movement Kinematics

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ABSTRACT

In recent years, virtual reality (VR) technology has shown promise as a means of delivering rehabilitative care to restore arm function in stroke patients. At the same time, limitations of traditional clinical scales for measuring arm function recovery have led to the more widespread use of *kinematic metrics*. These metrics quantify useful properties of patients' movements using motion tracking data captured while the patient performs different types of assessment tasks. Given modern consumer VR systems already collect the data needed to calculate many common kinematic metrics, these systems could eventually be used to both deliver stroke rehabilitation programs and administer kinematic assessments to monitor patients' recovery. However, it is not yet clear how the properties of VR-based assessment tasks may systematically impact the values of kinematic metrics used to assess arm function post-stroke. To begin addressing this question, we examined the influence of two task properties (movement direction and hand dominance) on a set of 10 kinematic metrics during a discrete reaching task performed by healthy participants using an Oculus Quest 2 VR headset. Our findings indicate that all 10 metrics were significantly impacted by these task properties, confirming that kinematic metrics captured in this context are sensitive to task properties. Our results also provide an initial account of how the metrics were influenced by each task property and highlight needs for future work to further understand the influence of assessment task properties on VR-based kinematic assessments.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality; Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Virtual Reality, Movement Kinematics, Motor Control, Stroke Rehabilitation

ACM Reference Format:

Logan D. Clark and Sara L. Riggs. 2022. VR-Based Kinematic Assessments: Examining the Effects of Task Properties on Arm Movement Kinematics. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*



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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA
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ACM ISBN 978-1-4503-9156-6/22/04.
<https://doi.org/10.1145/3491101.3519841>

(CHI '22 Extended Abstracts), April 29–May 05, 2022, New Orleans, LA, USA.
ACM, New York, NY, USA, ?? pages. <https://doi.org/10.1145/3491101.3519841>

1 INTRODUCTION

Stroke is a major cause of long-term disability that often causes severe arm function impairment [32, 36]. In recent years, virtual reality (VR) technology has shown promise as a means of delivering rehabilitative care to restore arm function in recovering stroke patients, partly by making rehabilitative exercises more engaging to increase the intensity and number of repetitions that patients perform [11, 15, 22, 28, 29, 32, 34]. The recent emergence of affordable all-in-one VR systems like the Oculus Quest could make VR-based rehabilitation programs more accessible and eventually allow patients to receive home-based VR therapy as part of stroke telerehabilitation programs [41, 48, 51].

In parallel, concerns regarding limitations of traditional clinical scales for measuring arm function recovery in stroke patients have led to more widespread use of *kinematic metrics* [36, 49]. These metrics quantify meaningful properties of a person's movements (e.g., speed, efficiency, smoothness) using motion tracking data captured while the person performs assessment tasks involving different types of arm movements [24, 36, 52]. There is evidence that compared to traditional clinical scales, kinematic metrics can be more amenable to parametric statistical analysis [8], and sensitive enough to detect small changes in motor function and provide a more complete picture of a patient's motor recovery [10]. These findings have culminated in recent expert recommendations calling for kinematic measures to be used alongside traditional clinical scales for assessing arm function in stroke patients [24, 42].

Interestingly, modern VR systems already collect the motion capture data needed to calculate many of the kinematic metrics that have been used to quantify post-stroke arm function. This means that VR systems could potentially be used to both (1) deliver stroke rehabilitation programs [41] and (2) administer kinematic assessments to monitor patients' progress [20]. This could provide new tools for doctors to remotely assess patients' arm function as part of stroke telerehabilitation programs, while also allowing kinematic assessments in clinical settings to be administered using more accessible and affordable hardware. To-date, this possibility has been considered for both screen-based VR systems [53, 54] and HMD-based VR-systems like the HTC Vive [20] and Oculus Rift [40]. This concept is further bolstered by recent findings suggesting that kinematic metrics derived using data from VR controllers can provide results similar to those obtained from traditional motion tracking systems (e.g., OptiTrack; [38]), and that kinematic assessments performed in stereoscopic virtual environments can index some aspects of arm function recovery in stroke patients [18, 19].

One known limitation of kinematic metrics is that their values can depend on the movement task during which they are measured [36]. In clinical research, where kinematic metrics have been used to measure the efficacy of new treatment techniques, the task-dependence of kinematic metrics has motivated the development of various standardized assessment tasks to improve the comparability of results across studies [24, 36]. However, there has been limited work examining *how* kinematic metrics are affected by different properties of assessment tasks, particularly in the unique perceptual conditions of a VR environment. For example, movements performed in different directions in 3D space (i.e., different *movement directions*) could yield different kinematic outcomes due to direction-dependent differences in motor noise [1, 2, 12–14], or limb stiffness [26, 27], as well as differences across 3D space in the reliability of the sensory information available to inform movements [2, 44]. There is also evidence to suggest that the kinematic properties of arm movements may vary based on whether movement is performed with the dominant or non-dominant arm (i.e., *hand dominance*; [55]), possibly because different control processes are more heavily involved in executing movements with each arm [46, 47].

One recent study aimed at comparing the kinematic properties of reaching movements performed in the real world and VR did find that some kinematic metrics (v_{peak} , MT , IOC ; Table 1) were not significantly influenced by *movement direction* or *hand dominance* in the context of a reach-to-grasp task performed in VR [3]. However, this study only examined movements performed away from the body to targets at different positions in the vertical plane, and it considered only three kinematic metrics. It remains unclear if differences may emerge across a broader range of movement directions, and for a more diverse set of kinematic metrics. This information could be useful for interpreting kinematic results from existing assessment tasks as they become adapted to VR (e.g., [40]) by providing a baseline understanding of how the properties of a given assessment task should be expected to impact kinematic results in VR.

To begin addressing this need, we examined the influence of two task properties—*movement direction* and *hand dominance*—on a battery of 10 kinematic metrics recorded during a discrete reaching task performed by healthy participants in VR. Kinematic metrics were selected from a systematic review [49], with the goal of including both the most frequently used metrics and some newer metrics that show promising initial results. We focused on a discrete reaching task rather than other task types (e.g., reach-to-grasp) because: (a) there is evidence that kinematic metrics captured during a VR version of this task can index arm function in stroke patients [18, 19], and (b) discrete goal-directed reaching movements make up a constituent part of many other assessment tasks that may eventually be adapted to VR, such as the drinking task [37], the box-and-blocks test [9, 40], and various activities of daily living [33]. Using healthy participants allowed us to isolate the effects of task properties on each kinematic metric from the effects of motor function impairment.

2 METHODS

2.1 Participants and Materials

Twenty healthy adults were recruited from the undergraduate, graduate student, and employee population at the University of Virginia (12 male, mean age = 23.6, range = 18–31). Participants had normal or corrected-to-normal vision and all expressed a strong right-hand preference, with scores greater than 40 ($M = 78.95$, $SD = 14.35$) on the Edinburgh Handedness Inventory [39]. Five participants reported some previous experience using a consumer VR headset.

The study was performed using an Oculus Quest 2 head-mounted display and Oculus Touch controllers, running custom software built in Unity. The experiment was performed in a 2.90m by 2.38m room. To minimize any effects of posture adjustments on arm movement kinematics, participants remained seated in a comfortable stationary chair that was secured to the floor. The experimenter sat at a desk to the side of participants and monitored their performance. Interpupillary distance (IPD) was measured for each participant using a medical grade pupillometer (Essilor Instruments, Model X81705), and the VR headset was adjusted to match each participant's IPD using the sizing recommendations provided by Oculus.

2.2 Experimental Task

The experimental task was a discrete point-to-point reaching task in which participants used a virtual hand mapped to the movement of the handheld controllers to reach to targets at different locations in 3D space (Figure 1). Participants began each trial in this task by moving their virtual hand until the tip of the index finger touched the yellow “start sphere.” Participants then pressed a specified button on the controller (“A” button for right hand movements, and “X” button for left) to select the start sphere while keeping their fingertip inside the start sphere. After a random interval (min = 0.5s, max = 1.5s), the start sphere disappeared and a red “target sphere” appeared somewhere in the environment. Participants then moved their virtual hand to select the target sphere. When the fingertip of the virtual hand touched the target sphere, the sphere's color darkened to indicate that it could be selected. Participants then pressed the appropriate button on the controller to select the target sphere. Once a target sphere was selected it disappeared, and the start sphere reappeared to allow participants to begin the next trial. Participants were instructed to select each target sphere “as quickly and accurately as possible” and only to press the button to select a target once they were sure that they had reached the target. These instructions were designed to elicit aimed reaching movements that reflect the constraints of typical real-world reaching movements, in which participants must accurately specify both the direction and extent of the movement (e.g., [7, 30, 50]). Participants were instructed to rest as needed between trials to minimize any effects of fatigue.

The position and size of objects were defined using Unity units (1 unit $\approx$ 1 meter). At the beginning of each session, the experimenter calibrated the VR headset to ensure that the location of targets relative to participants would remain consistent across all participants and sessions. All the spheres in this study were 0.05 units in diameter. Start spheres could either appear 0.25 units to the left of the participant's body midline (i.e., in the “left hemisphere”),

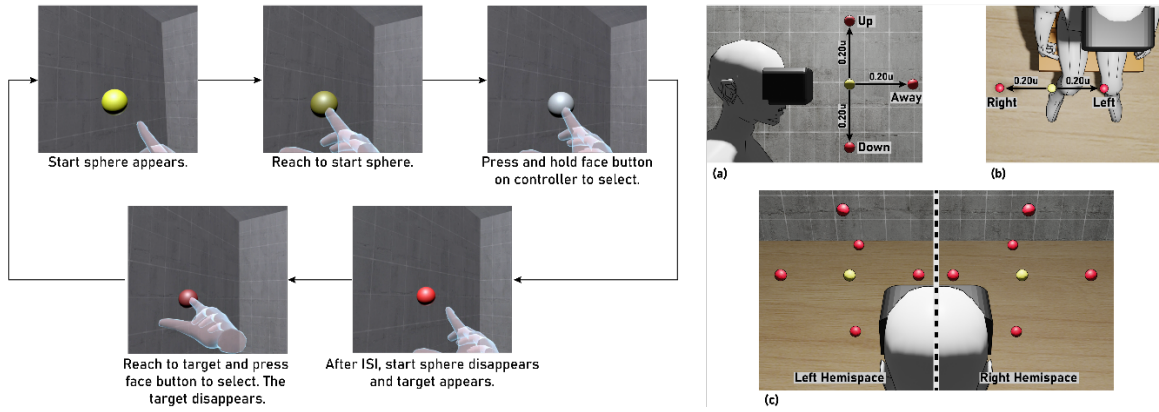


Figure 1: On the left, the sequence of events for the reaching task used in the present study. On the right, the potential start and target sphere locations, viewed from the side (a), the top (b), and from behind the participant (c). (*Note: All distances are in Unity units).

or 0.25 units to the right of the body midline (i.e., in the “right hemisphere”). In both cases, start spheres were positioned at participants’ eye level, which was fixed at 1.00 units above the virtual floor, and 0.25 units away from participants along the depth axis (Figure 1). Targets could appear 0.2 units away from their associated start sphere in one of five different directions (i.e., “up”, “down”, “left”, “right”, “away”; Figure 1). These same locations were used to position targets relative to both potential start sphere locations, yielding the full set of potential target locations shown in Figure 1. An additional set of 12 other target locations involving different movement distances were also included in this study and will be explored in future work. Here, we focus on a subset of 5 equidistant targets and average results across movements in both hemispaces to isolate the effects of movement direction and hand dominance on when movement distance is held constant.

2.3 Procedure

Participants were first introduced to the experimental task using a set of target locations that were not used during the experimental sessions. Participants then completed four experimental sessions. Within each session, the location of the start sphere (i.e., the *interaction hemisphere*; “left side”, “right side”) and the *hand used* to perform the reaching movements (“left hand”, “right hand”) remained constant. The levels of these two factors were crossed orthogonally to produce the four experimental sessions. Session order was counterbalanced across participants using a Latin square design. The *movement direction* in each trial was randomized, with the constraint that each potential movement direction occurred 10 times in each session. Participants completed 170 trials in each of the four sessions, for a total of 680 trials per participant. Of these, 200 trials per participant involved movements to the target locations of interest in the present work (Figure 1). For the present analysis, where we focus on the effects of *hand dominance* and *movement direction* for movements in both hemispaces, this resulted in a 2 (*hand used*; dominant right/non-dominant left) $\times$ 5 (*movement direction*; up, down, left, right, away) repeated measures design.

2.4 Data Analysis

Data preprocessing was performed using MATLAB R2021a [31]. The experimental software captured the x -, y -, and z -position of the virtual fingertip at a nominal sampling rate of 90 Hz. Missing position data were interpolated using spline interpolation, and the data were resampled to a constant 90 Hz sampling rate. The position data were then filtered using a 2nd order Butterworth filter with settings appropriate for calculating each kinematic metric (i.e., half amplitude cutoff at 20 Hz for SPARC and LDLJ, and 8 Hz for all other metrics [16, 23]). Filtered position data were then used to calculate the cumulative Euclidean distance travelled by the virtual fingertip at each time point during each session. The resulting profiles were differentiated to derive the velocity and acceleration profiles used to calculate kinematic metrics [16]. Finally, each metric was calculated for each trial using the procedures described in its original publication (Table 1).

Of the 4000 total trials, two (0.05%) contained evidence of tracking loss, which appeared as a flatline in the velocity curve followed by impossibly large velocity values (>10 units/second) as the finger appeared to travel a large distance instantaneously when tracking resumed. In addition, 95 trials (2.4%) contained evidence of target selection errors, whereby participants either paused for a long time during the movement (i.e., movement time greater than $3 \times$ IQR above the third quartile) or failed to select the target with their initial reach (i.e., the button was pressed more than once during a trial). Movement data from these two sets of trials would not accurately reflect the kinematics of typical goal-directed reaching movements, so these trials were removed from the dataset and the remaining 3903 trials were submitted for further analysis. The effects of *hand dominance* (“hand”) and *movement direction* (“direction”) on the values of each kinematic metric were assessed using separate multilevel linear models [25] implemented using the lme4 package [5] in R version 4.0.5 [43]. The models for each kinematic metric were fitted using full maximum likelihood estimation following a bottom-up procedure, beginning with a model containing only random intercepts for participants and adding fixed effects one at a time for the two predictor variables and their interaction.

Table 1: The kinematic metrics examined in the current study.

Metric Name	Notation	Property	Description
Peak Velocity	v_{peak}	Speed	Maximum speed (units/s) reached by the virtual fingertip between the start and end of the movement. Higher values indicate faster movements.
Movement Time	MT	Efficiency	Time elapsed (in seconds) between the beginning and end of the movement. Higher values indicate less efficient movements.
Percent Time to Peak Velocity	$PTPV$	Planning	The percentage of MT that has elapsed when v_{peak} occurs. Higher values indicate more effectively planned movements.
Mean Velocity	v_{mean}	Speed	The mean speed (units/s) of the virtual fingertip between the beginning and end of the movement. Higher values indicate faster movements.
Normalized Mean Velocity	v_{norm}	Smoothness	The ratio of v_{mean} to v_{peak} . Higher values indicate smoother movements.
Number of Velocity Peaks	n_{peaks}	Smoothness	The number of local maxima in the velocity profile for a movement [4]. Higher values indicate less smooth movements.
Index of Curvature	IOC	Efficiency	The ratio of the distance traveled by the fingertip to the straight-line distance between the start sphere and the target. Higher values indicate less efficient movements.
Initial Direction Error	IDE	Planning	The angular deviation (in degrees) between (1) the 3D vector connecting the start sphere to the target and (2) the vector connecting the start sphere to the finger's position when v_{peak} occurs. Higher values indicate less effectively planned movements.
Log Dimensionless Jerk	$LDLJ$	Smoothness	The integral of jerk (the third derivative of position) during the movement, with a correction applied to prevent the value from varying with the amplitude and duration of the movement [4]. This metric is negated so that higher values indicate smoother movements.
Spectral Arc Length	$SPARC$	Smoothness	The arc length of the frequency spectrum obtained by applying a Fourier transform to the velocity profile for a movement, within a dynamically defined frequency range [4]. This metric is negated so that higher values indicate smoother movements.

Likelihood ratio tests were used to determine if adding each factor significantly improved the fit. The p -values for these tests were adjusted using Holm's step-down procedure [17] to control the familywise error rate at 0.05, correcting for multiple comparisons. This allowed us to be confident in the results of each individual likelihood ratio test. Significant effects were further explored using Bonferroni-corrected post-hoc comparisons.

3 RESULTS

Likelihood ratio tests revealed that all 10 metrics were significantly influenced by some combination of *hand dominance* ("hand") and *movement direction* ("direction") in this VR-based discrete reaching task (Figure 2). We explore these effects in more detail below. We note that our sample size and multilevel analysis approach provided sufficient statistical power to detect relatively small effects. Therefore, we report all significant effects and their associated marginal means and leave it to the reader to determine if a given effect is large enough to achieve practical significance in their specific use context.

Significant main effects of *hand* on kinematic metric values generally pointed to performance benefits for movements performed with the right hand (RH; dominant hand) compared to the left hand (LH; non-dominant hand). Specifically, movements with the right hand exhibited significantly higher $LDLJ$ values (RH: $M = -10.2$,

$SE = 0.186$; LH: $M = -10.4$, $SE = 0.186$; $p < .001$) and significantly lower n_{peaks} values (RH: $M = 5.32$, $SE = 0.266$; LH: $M = 5.61$, $SE = 0.266$; $p < .001$), indicating smoother movements. Movements with the right hand also exhibited significantly higher v_{mean} (RH: $M = 0.291$, $SE = 0.01$; LH: $M = 0.280$, $SE = 0.01$; $p < .001$) and $PTPV$ values (RH: $M = 54.6$, $SE = 1.40$; LH: $M = 53.0$, $SE = 1.40$; $p < .001$), and significantly lower MT values (RH: $M = 0.807$, $SE = 0.026$; LH: $M = 0.840$, $SE = 0.026$; $p < .001$), indicating improved speed, planning, and efficiency. There were also significant main effects of *hand* on v_{norm} and v_{peak} , but these effects were subsumed by a significant interaction (Figure 2).

Significant main effects of *direction* on kinematic metric values reflected several different patterns (Figure 3). First, movements in the down direction exhibited significantly higher MT values ($M = 0.860$, $SE = 0.027$) than movements in the other four directions (Away: $M = 0.811$, $SE = 0.027$; Up: $M = 0.813$, $SE = 0.027$; Left: $M = 0.814$, $SE = 0.027$; Right: $M = 0.819$, $SE = 0.027$; $p < .001$ for all comparisons), and higher IOC values ($M = 1.55$, $SE = 0.017$) than movements in the away ($M = 1.48$, $SE = 0.017$) and right directions ($M = 1.50$, $SE = 0.017$;), indicating a trend of decreased movement efficiency for downward movements. Second, $SPARC$ values were significantly higher for movements in the up ($M = -1.81$, $SE = 0.013$) and down ($M = -1.80$, $SE = 0.013$) directions compared to the away ($M = -1.85$, $SE = 0.013$) and right directions ($M = -1.84$, $SE = 0.013$; $p < .001$ for both comparisons), indicating a smoothness benefit

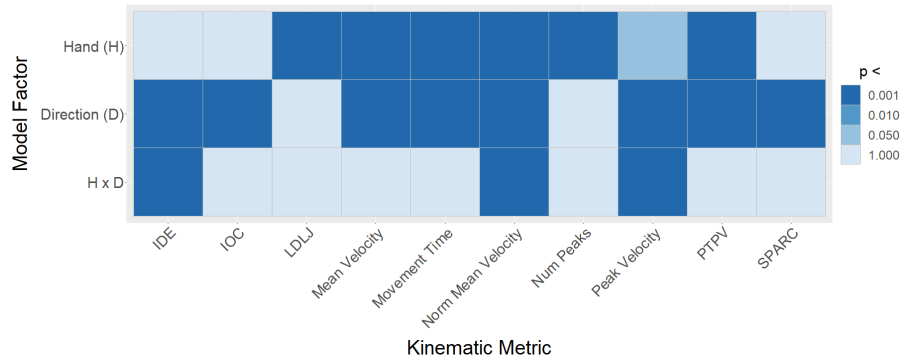


Figure 2: Summary of the Holm-corrected p-values from all likelihood ratio tests. The color in each cell indicates the extent to which each Model Factor—i.e., two independent variables [hand (H), direction (D)] and their interaction (H x D)—exerted significant effects on each Kinematic Metric (e.g., IDE, IOC, etc.).

for movements in these directions. Finally, there was a significant trend of direction-dependent differences in *PTPV* values, whereby *PTPV* was largest for movements in the away direction ($M = 57.0$, $SE = 1.43$), followed by the up ($M = 55.3$, $SE = 1.43$), down ($M = 53.3$, $SE = 1.43$), and both the left and right directions (Left: $M = 51.7$, $SE = 1.43$; Right: $M = 51.8$, $SE = 1.43$), indicating significant differences in movement planning effectiveness ($p < .001$ for all comparisons). There were also significant main effects of *direction* on *IDE*, v_{norm} and v_{peak} , but these effects were subsumed by a significant interaction (Figure 2).

We also observed significant *hand* $\times$ *direction* interaction effects for three metrics, indicating that the effect of *direction* on kinematic metric values varied based on which *hand* was used to complete the movements. The most prominent pattern underlying these effects involved performance differences for movements performed *toward* the side of the hand being used (i.e., movements to the left with the left hand, or movements to the right with the right hand) compared to movements *away* from the side of the hand being used. For example, for movements performed with the left hand, v_{peak} values were significantly higher for movements to the left ($M = 0.808$, $SE = 0.027$) than for movements to the right ($M = 0.732$, $SE = 0.027$; $p < .001$), indicating improved movement speed for movements in the left direction. However, for movements performed with the right hand, v_{peak} values were significantly higher for movements performed to the right ($M = 0.810$, $SE = 0.027$) than for movements to the left ($M = 0.775$, $SE = 0.027$; $p < .05$). Analogous patterns were observed for *IDE* and v_{norm} , whereby *IDE* values were significantly lower (indicating more effective movement planning) and v_{norm} values were significantly lower (indicating less smooth movements) for movements performed *toward* the side of the hand being used than for movements performed *away* from the side of the hand being used ($p < .001$ for all comparisons).

4 DISCUSSION

Virtual reality has already shown promise as a means of delivering rehabilitative care for stroke patients [11, 15], and modern consumer VR systems could soon offer a flexible and accessible platform for administering kinematic arm function assessments to monitor patients' recovery. Although it is known that kinematic

metrics can be influenced by the assessment task during which they are measured [24, 36], little work to-date has examined *how* specific properties of an assessment task influence kinematic metrics used to assess arm function post-stroke, particularly for assessment tasks performed in VR. The present study explored this question for a VR-based discrete reaching task [18, 19] by examining the impact of two task properties, *movement direction* and *hand dominance*, on a set of 10 kinematic metrics captured while healthy participants performed this task. Results revealed that all 10 metrics were significantly influenced by some combination of these two properties, confirming that kinematic metrics captured in this context can be sensitive to task properties, not just motor impairment. Importantly, we also provide a novel account of how *movement direction* and *hand dominance* influence the values of each kinematic metric in this context. These initial findings suggest that researchers and practitioners should account for the influence of task properties when using kinematic metrics to quantify arm function in VR environments. This could involve: (a) considering any known effects of task properties on kinematic metrics when interpreting the results of kinematic assessments, or (b) developing and using kinematic metrics that are more robust to the effects of task properties.

Building from this initial contribution, additional work is needed to further explore how other task properties impact a broader set of kinematic metrics and assessment tasks. To this end, forthcoming work from our laboratory will extend this initial investigation to explore how other properties of a VR-based discrete reaching task may interact with *movement direction* and *hand dominance* to influence an even broader set of kinematic metrics, including emerging metrics based on submovement decomposition analyses [45]. Furthermore, to explore how the effects of task properties on kinematic metrics may vary across individuals, these studies will be replicated with different participants, and random effects models will be used to study between-participant variability in the effects of task properties on kinematic measures.

By providing a foundational understanding of how assessment task properties impact kinematic measures even in the absence of motor impairment, the findings reported here and in forthcoming work can help researchers and clinicians to distinguish task-related

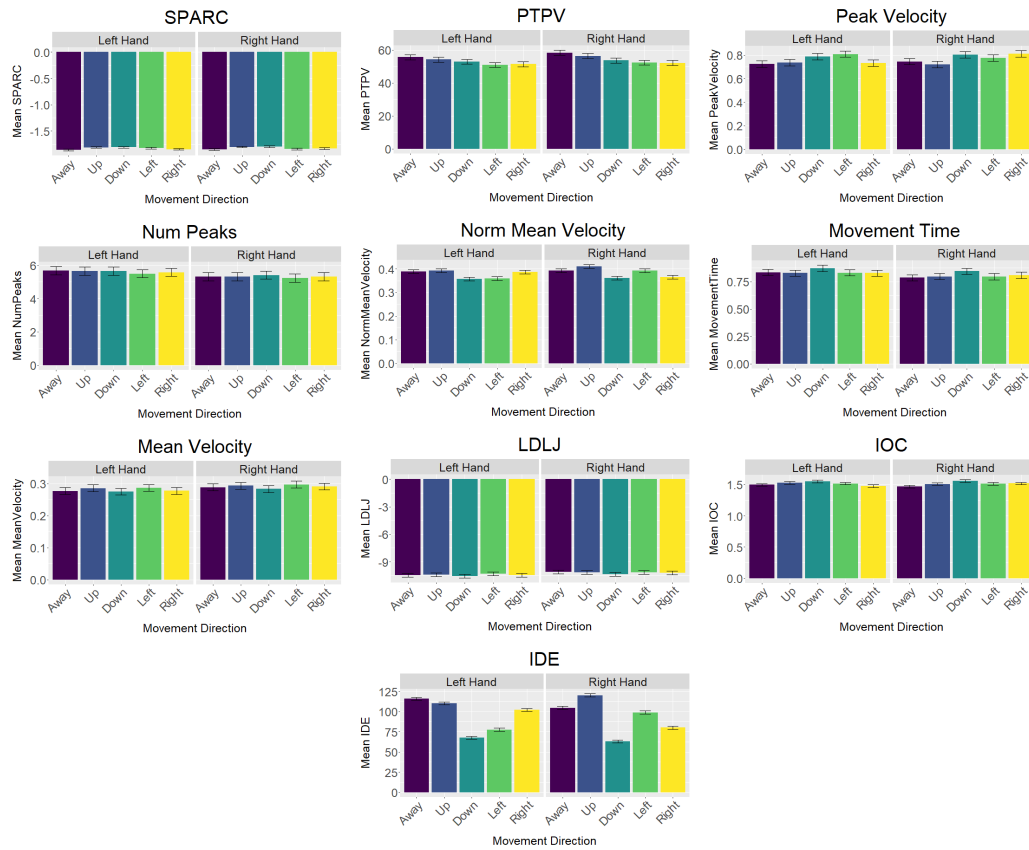


Figure 3: Bar plots displaying the mean values of each kinematic metric in each experimental condition. Error bars reflect the estimated standard error of the mean. Because *SPARC* and *LDLJ* always take negative values, the bar plots for these metrics are inverted (i.e., zero is at the top of the *y*-axis rather than the bottom).

differences in kinematic metrics from lesion-related movement quality deficits that should be addressed in therapeutic interventions. Although our focus here is on kinematic assessments that involve discrete reaching tasks performed in VR (e.g., [18, 19]), this understanding could also be useful for interpreting kinematic results from other reaching-based assessment tasks as they are adapted to VR (e.g., [40]). In the long term, these findings could also inform other applications of kinematic measures in VR environments, such as monitoring progress during motor skill training [35] or identifying usability issues in VR interfaces [6, 21].

ACKNOWLEDGMENTS

Logan Clark was supported by a Meta PhD Fellowship. The VR software used in this study was developed by Peiyu Zhang and Jeffrey Richbart. We would also like to thank the four anonymous reviewers whose comments and suggestions helped to improve this manuscript.

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